ch22 AC and Domestic Electricity

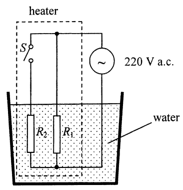
Items:

Q8
Q8
Q8
Q8
Q8
Q8

Q8

8. In the circuit shown in Figure 8.1, resistors R_1 and R_2 represent the heating elements in a heater using mains supply. Both resistors are immersed in water.

Figure 8.1



The heater can be operated in two modes, namely, heating and keeping warm, and it is controlled by the switch S. The power consumed by the heater in the heating mode is 550 W and in the mode of keeping warm is 88 W. The mains voltage is 220 V a.c.

(a) In which mode is the heater operating when switch S is open ? (1 mark)

....

(b) Find the resistance of R_1. (2 marks)

....

....

....

....

....

....

(c) When switch S is closed, calculate the current passing through resistor R_2. (3 marks)

....

....

....

....

....

....

....

....

Answers written in the margins will not be marked.

*(d) What is the *peak value* of the sinusoidal current flowing through the heater when switch S is closed ? (2 marks)

....

....

....

....

....

....

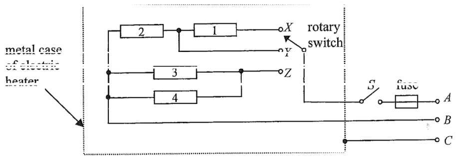
....

Answers written in the margins will not be marked.

Q8

8. Figure 8.1 shows the schematic diagram of an electric heater consisting of four identical heating elements, each having a rated value of '500 W 220 V'. A user can use the rotary switch to select one of the three modes of operation X, Y, Z. Wires A, B, C from the heater are connected to the 220 V a.c. mains via a 3-pin plug.

Figure 8.1



(a) Find the resistance R of a heating element. (1 mark)

(b) What is the total power dissipation when mode X is selected ? Assume that the resistance of each heating element remains unchanged. (2 marks)

(c) Without the need of calculations, explain which mode of operation has the largest total power dissipation. (2 marks)

(d) (i) If fuses 3 A, 5 A and 13 A are available, determine which will be the most suitable to limit an excess current. Show your work. (3 marks)

(ii) A student claims that since a.c. is used for the heater, the switch S can be installed in either wire A or wire B. Comment on this claim. (2 marks)

(iii) If a fault resulted in the live wire having contact with the metal case of the heater, which wire, A, B or C, could prevent an electric shock of a person touched the case of the heater ? Explain. (2 marks)

Q8

8. (a) Figure 8.1 shows the schematic diagram of an electric heater which can operate in two modes, namely 'heating' and 'keeping warm'. The heating elements of resistances $4R$ and R are connected to the mains supply via a 3-way switch with its two poles tied together. That is, both poles can be connected to one of the three pairs of terminals X, Y or Z.

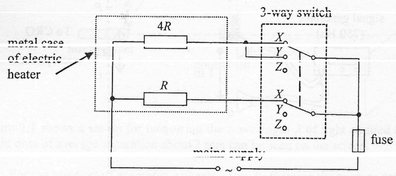


Figure 8.1

(i) To which pairs of terminals, X, Y or Z, should the switch connect to when the heater is in 'heating' mode ? (1 mark)

The power consumed by the heater in 'heating' mode is 800 W.

(ii) Calculate the current drawn from the 220 V mains supply when the heater is in 'heating' mode. (2 marks)

(iii) Find the power consumed by the heater in the mode of 'keeping warm'. (3 marks)

Answers written in the margins will not be marked.

(b) Figure 8.2 shows a simplified domestic circuit connected to an electrical appliance via a fuse, a meter M, a residual current circuit breaker (RCCB) and a switch.

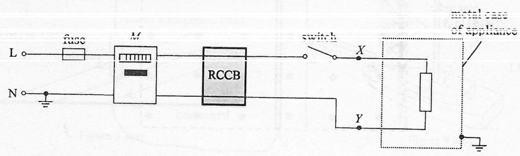


Figure 8.2

(i) What physical quantity does the meter M record ? (1 mark)

(ii) An RCCB is a kind of safety device that cuts off the supply automatically whenever there is a small difference between the currents in the live (L) and neutral (N) wires. State, in each of the following situations, which device(s) will respond (i.e. the fuse blows and/or the RCCB cuts off the supply).

(1) A short circuit occurs between points X and Y. (1 mark)

(2) A short circuit occurs between point Y and the metal case of the appliance. (1 mark)

Q8

8. Figure 8.1 shows a household electrical wiring circuit. The mains cable (containing live wire L and neutral wire N) is connected to a consumer unit via a kilowatt-hour meter M. At the consumer unit, the wires branch out into a number of parallel circuits.

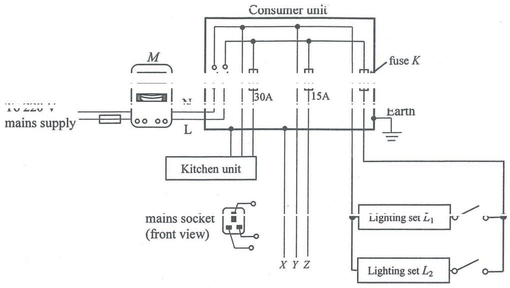


Figure 8.1

(a) Indicate on Figure 8.1 how the mains socket should be connected to wires X, Y and Z. (1 mark)

(b) Lighting sets L_1 and L_2 of power ratings 300 W and 450 W respectively are connected in parallel to the branch with fuse K.

(i) State one advantage of connecting L_1 and L_2 in parallel instead of in series to the branch. (1 mark)

....

....

....

(ii) If fuses marked 3 A, 5 A, 10 A and 13 A are available, which one is the most suitable to be fuse K? Explain your choice. (3 marks)

....

....

....

....

....

....

....

Answers written in the margins will not be marked.

(c) The kitchen unit includes the following electrical appliances:

	rating	effective time of operation at rated value per day
a refrigerator	220 V, 500 W	8 hours
an electric kettle	220 V, 2000 W	0.5 hour
an induction cooker	220 V, 3000 W	2 hours

How much should be paid per day to run these appliances if 1 kW h of electrical energy costs $0.9 ? (3 marks)

....

....

....

....

....

....

....

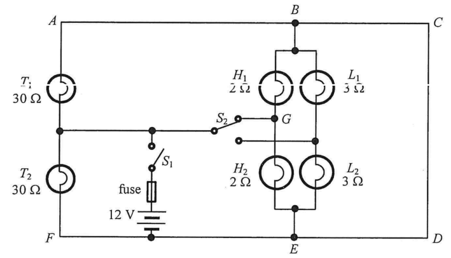
....

Answers written in the margins will not be marked.

Q8

8. Figure 8.1 shows a simplified circuit of the lighting system of a car. Each of the taillights (T_1, T_2), high-beam headlights (H_1, H_2) and low-beam headlights (L_1, L_2) has resistance 30 Ω, 2 Ω and 3 Ω respectively. The internal resistance of the 12 V battery and the resistance of the fuse are negligible.

Figure 8.1



When switch S_1 is closed and switch S_2 is set at the position shown in Figure 8.1, only T_1 and T_2 as well as H_1 and H_2 are lit. The current drawn from the battery is at a maximum in this setting.

(a) Explain why L_1 and L_2 are **not lit**. (1 mark)

....

....

....

....

....

(b) (i) What is the potential difference across the taillight T_2 ? (1 mark)

....

(ii) Indicate on Figure 8.1 the direction of current in each of the branches AB, GB and BC. Which branch carries the largest current ? (3 marks)

....

Answers written in the margins will not be marked.

(c) Calculate the power delivered by the battery and show that the equivalent resistance of the circuit is slightly less than 1 Ω in this setting. (4 marks)

....

....

....

....

....

....

....

....

....

....

....

....

....

(d) Based on your answer in (c), explain whether a fuse rating of 15 A is suitable for this circuit or not. (2 marks)

....

....

....

....

....

....

....

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

Q8

Please stick the barcode label her

8. Figure 8.1 shows an earth leakage circuit breaker (漏電斷路器) installed in a domestic circuit. The live and the neutral wires pass through the centre of a soft iron ring of mean radius 1 cm. A 100-turn coil C with cross-section area 0.8 cm^2 is wound on the ring.

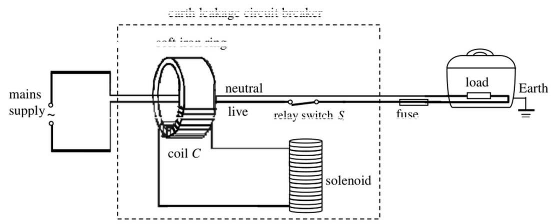


Figure 8.1

In case of an earth leakage in the domestic circuit such that the current in the neutral wire and the live wire differ by 0.5 A or more, the relay switch S opens and disconnects the mains supply. To reconnect the supply, S has to be reset manually.

(a) Explain why S opens when there is a leakage current of 0.5 A from the load to the Earth. (3 marks)

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

(b) Calculate the magnetic field B through coil C when there is a leakage current of 0.5 A from the load to the Earth. The magnetic field B due to a current-carrying conductor is 1500 times larger in soft iron. (2 marks)

(c) Electrical appliances are usually equipped with fuses. When a short circuit occurs between the live and neutral wires, the fuse blows but the earth leakage circuit breaker does not operate. Explain these observations. (2 marks)

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.